DA2402 — Data Curation and Visualization

Tier	A
Phase (order)	3
Do this first	DA1301
Estimated effort	8 weeks
Unlocks	DA3406, DA5500W

Roadmap

1. Collection, cleaning (missing values, outliers), wrangling (groupby, pivot, merge, time series).
2. Visual design principles; Matplotlib/Seaborn/Plotly dashboards.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Data Science for Engineers | IIT Madras (50 videos). NPTEL *Data Science for Engineers* (IIT Madras): data handling, stats, regression, clustering, in R.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Pandas Tutorials (11 videos). Corey Schafer *Pandas Tutorials*: loading, cleaning, grouping and reshaping real data.
- Matplotlib Tutorials (10 videos). Corey Schafer *Matplotlib Tutorials*.
- Data Analysis with Python Course (6 videos). freeCodeCamp *Data Analysis with Python*.
- Building Data Apps in Python - Plotly Dash (7 videos). Plotly *Building Data Apps in Python (Dash)* (official): interactive dashboards.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–11	R basics: data types, data frames, functions, control flow, plotting. (You may do the same work in Python/pandas.)
2	12–18	Linear algebra for data science.
3	19–22	Statistical modelling, random variables, sample statistics, hypothesis testing.
4	23–28	Optimisation: unconstrained, gradient descent, equality and inequality constraints.

Week	Lectures	What to do
5	29–40	Data science process, linear regression, model assessment, diagnostics, cross-validation, multiple regression. Do a regression project on a real dataset.
6	41–50	Classification: logistic regression, performance measures, kNN, k-means. Do a Kaggle classification task end to end.

▼ **All lectures in *Data Science for Engineers* | *IIT Madras* (50 videos)**

1. Data science for engineers Course philosophy and expectation
2. Introduction to R
3. Introduction to R (Continued)
4. Variables and datatypes in R
5. Data frames
6. Recasting and joining of dataframes
7. Arithmetic,Logical and Matrix operations in R
8. Advanced programming in R : Functions
9. Advanced Programming in R : Functions (Continued)
10. Control structures
11. Data visualization in R Basic graphics
12. Linear Algebra for Data science
13. Solving Linear Equations
14. Solving Linear Equations (Continued)
15. Linear Algebra - Distance,Hyperplanes and Halfspaces,Eigenvalues,Eigenvectors
16. Linear Algebra - Distance,Hyperplanes and Halfspaces,Eigenvalues,Eigenvectors (Continued 1)
17. Linear Algebra - Distance,Hyperplanes and Halfspaces,Eigenvalues,Eigenvectors (Continued 2)
18. Linear Algebra - Distance,Hyperplanes and Halfspaces,Eigenvalues,Eigenvectors (Continued 3)
19. Statistical Modelling
20. Random Variables and Probability Mass/Density Functions
21. Sample Statistics
22. Hypotheses Testing
23. Optimization for Data Science
24. Unconstrained Multivariate Optimization
25. Unconstrained Multivariate Optimization (Continued)
26. Gradient (Steepest) Descent (OR) Learning Rule
27. Multivariate Optimization With Equality Constraints
28. Multivariate Optimization With Inequality Constraints
29. Introduction to Data Science
30. Solving Data Analysis Problems - A Guided Thought Process
31. Module : Predictive Modelling
32. Linear Regression
33. Model Assessment

34. Diagnostics to Improve Linear Model Fit
35. Simple Linear Regression Model Building
36. Simple Linear Regression Model Assessment
37. Simple Linear Regression Model Assessment (Continued)
38. Muliple Linear Regression
39. Cross Validation
40. Multiple Linear Regression Modelling Building and Selection
41. Classification
42. Logisitic Regression
43. Logisitic Regression (Continued)
44. Performance Measures
45. Logisitic Regression Implementation in R
46. K - Nearest Neighbors (kNN)
47. K - Nearest Neighbors implementation in R
48. K - means Clustering
49. K - means implementation in R
50. Data Science for engineers - Summary

Books, notes and websites

- Book: McKinney, *Python for Data Analysis* (free online)
- Knaflic, *Storytelling with Data*
- pandas and Plotly docs; Kaggle *Data Cleaning* course
- ✅ Nearest NPTEL match: *Data Science for Engineers* (Rengasamy & Narasimhan, IIT Madras) — nptel.ac.in/courses/106106179. NPTEL has no dedicated data-visualization course, so this covers only part of it.

Practice / proof of understanding

Clean a messy public dataset and ship a 5-chart story (notebook + short report).

What your IITM course sheet says

From the PDF: DA2402

Official description: By the end of the course, students will be able to: • Describe and implement the steps involved in data curation (collection, cleaning &; wrangling) • Understand core principles of effective data visualization • Create static and interactive charts • Design and build interactive dashboards

Official syllabus (trimmed):

1. Module 1: Introduction to Data Curation: Importance of data curation Data acquisition and collection Data lifecycle management Data governance & compliance;
2. Module 2: Data cleaning methods: Types of data (quantitative, categorical, temporal etc) Common data quality issues Handling missing data Outlier detection and treatment methods Handling other data inconsistencies
3. Module 3: Data wrangling techniques: Data summarization (Groupby, Pivot) Data reshaping (melting, stack- ing) Data integration from multiple sources Working with time series data
4. Module 4: Introduction to Data Visualization: Importance of data visualization Application across industries Types of charts Python visualization libraries overview (Matplotlib, plotly, seaborn) Introduction to grammar of graphics Types of analysis (univariate, bivariate, multivariate)
5. Module 5: Standard visualization charts: Charts for statistical analysis (box plots, histograms etc) Univariate & Bivariate analysis (Distribution analysis, Frequency analysis, Trending analysis)
6. Module 6: Advanced charts: Understanding multi-variate analysis Charts for Hierarchical analysis (stacked bars, Treemaps, Sunburst) Geographical analysis (Maps) Text analysis (word clouds) Visualizing time series data
7. Module 7: Building interactive dashboards using Python: Designing a dashboard Adding filters for inter- activity C

Official textbooks: 1. Iliinsky, N., & Steele, J. (2011). Designing data visualizations: Representing informational relationships. O'Reilly Media. 2. Knaflic, C. N. (2015). Storytelling with data: A data visualization guide for business professionals. Wiley. 3. McKinney, W. (2022). Python for data analysis: Data wrangling with Pandas, NumPy, and Jupyter (3rd ed.). O'Reilly Media.

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